

HONEYCHAIN

Farm-to-Jar Blockchain Traceability & Smart Beekeeping Platform

Smart India Hackathon 2026 | PS SIH26021 | Ministry of MSME | Theme: Smart Automation | Software | Team Vibranium - SRM IST Chennai

THE PROBLEM - INDIA'S HONEY TRUST DEFICIT

- **World #2 honey exporter:** USD 206 M exports FY25 (APEDA) - yet trade runs on distrust.
- **Purity crisis:** CSE 2020 probe found top brands failing NMR tests; syrup blending is systemic.
- **Honest beekeepers lose:** no provenance = commodity prices; export lots rejected abroad.
- **Gap today:** Madhukranti registers beekeepers (24.9 lakh colonies) but cannot verify a retail jar.

Capability	Madhukranti	Ours
Registration & IDs	Yes	Integrates
Tamper-proof custody	No	YES
Consumer jar-QR check	No	YES
Smart hive app	No	YES
AI adulteration flags	No	YES

OUR SOLUTION

"Every honey jar carries a tamper-proof digital identity from apiary to shelf - verifiable by anyone with a single QR scan."

1. **Beekeeper App (Flutter)** - offline-first, Hindi/Tamil voice UI, GPS + photo harvest logs
2. **Blockchain Chain-of-Custody (Hyperledger Fabric)** - each handover signed by a different org key; immutable trail
3. **AI Anomaly Engine** - weight-physics checks expose syrup addition; disease + yield alerts
4. **FPO / Processor Console** - batch aggregation, payout transparency, compliance exports
5. **Consumer Trust Portal** - scan jar QR -> farm map, farmer profile, lab certificate, journey

HOW IT WORKS - ONE BATCH'S JOURNEY

1. Ravi harvests 42 kg mustard honey -> app creates Batch B-1042 with GPS, photo, date (works offline).
2. Collection center weighs 42 kg, matches manifest, signs custody -> transaction locked forever.
3. Processor shows 55 kg output -> AI physics check flags impossible gain = syrup added.
4. NMR lab certificate attaches to batch lineage via the laboratory's own key.
5. Buyer scans jar QR: farm map, Ravi's story, certificates, every hop dated. Trust the jar, not the label.

Why blockchain? A database is editable by its owner. Here every hop is signed by a different organisation's key - nobody, not even we or an FPO, can rewrite history.

IMPACT AT SCALE

Rs 700 Cr+ premium value unlocked per year at national scale	+15-25% income uplift per beekeeping household	24.9 Lakh colonies reachable via Madhukranti integration
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GOVERNMENT ALIGNMENT & WHAT WE ASK OF YOU

- **Policy fit:** executes NBHM objective (d) - blockchain traceability of honey sources - extends Madhukranti, complements KVIC Honey Mission.
- **Our ask:** (1) a line of expert review we may cite; (2) validation of field-workflow assumptions; (3) guidance to pilot with one TN beekeeper society/FPO.

Team Vibranium (6 members): AI/Full-stack · Mobile Development · Research, Documentation & Pitch
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